

Pangmiaomiao Zhang

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Ph.D. Candidate (Graduating May 2027) | Bioanalytical Assay Development | Nucleic Acid-Protein Interactions

QUALIFICATION

- Analytical chemist with expertise in bioanalytical assay development and protein-nucleic acid interactions, with hands-on experience in designing and optimizing biochemical and nucleic acid-based experimental workflows.
- Experienced in quantitative characterization of biomolecular interactions using HS-AFM, gel electrophoresis, mass photometry, fluorescence-based assays, HPLC-MS, and TEM, supported by quantitative data analysis and experimental troubleshooting.
- Demonstrated experience in method development, protocol optimization, instrument troubleshooting, mentoring, and interdisciplinary collaboration across molecular biology, biochemistry, structural biology, and bioanalytical research.

EDUCATION

The University of Texas at Austin , Ph.D. in Chemistry (GPA: 3.95/4.00)	Aug 2022-May 2027
Tsinghua University , M.Sc. in Chemistry (GPA: 3.87/4.00)	Sep 2019-Jun 2022
Southern University of Science and Technology , B.Sc. in Chemistry (GPA: 3.85/4.00)	Sep 2015-Jun 2019

RESEARCH EXPERIENCE

Graduate Researcher, Lin Bio-AFM Lab, UT Austin – Austin, TX	2023-Present
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Selected Research Projects

- Single-Molecule Visualization of G2L4 Reverse Transcriptase (RT)-Mediated DNA Repair**
 - Designed and prepared DNA oligonucleotide substrates with microhomology regions for DNA repair assays.
 - Designed and optimized biochemical and single-molecule assay workflows to characterize reverse transcriptase-DNA interactions, DNA repair intermediates and ligase activity.
 - Led to a first-author manuscript accepted in *Nature Communications*.
- Impact of Viral Glycosylation on SARS-CoV-2 Spike-Antibody Interactions** (*Manuscript in Preparation*)
 - Investigated how N- and O-linked glycosylation affects a N-terminal domain (NTD)-targeting antibody binding, complex formation, and aggregation of SARS-CoV-2 spike proteins.
 - Developed an integrated biochemical workflow using enzymatic deglycosylation, SDS-PAGE, mass photometry, HS-AFM and TEM to quantify changes in antibody binding, complex stoichiometry and aggregation.
- Structural and Membrane-Associated Characterization of SARS-CoV-2 Membrane (M) Protein**
 - Expressed and purified full-length SARS-CoV-2 membrane protein and optimized detergent solubilization, liposome reconstitution and sample-preparation workflows for downstream structural characterization using HS-AFM, TEM and cryogenic electron tomography.

Graduate Researcher, Tsinghua University – China	2019-2022
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- Fluorescence Sensor Arrays for Disease-Related Biomolecular Analysis**
 - Developed and optimized fluorescence probe-based bioanalytical assays to classify aggregation states of amyloid beta peptides and disease-associated urinary protein profiles, including sample preparation, probe selection, incubation conditions, and quantitative signal acquisition.
 - Established quantitative multivariate analysis workflows to distinguish aggregate states related to disease progression, characterize mixtures, and monitor reaction kinetics and pathways.
 - Applied the assay workflow to processed human urine samples in a clinical collaboration to distinguish healthy and kidney disease-associated protein profiles.

2. Detection of Histone Acetylation Related Enzymes Using Conjugated Polyelectrolytes

- Designed and developed polymer-peptide fluorescence probes for p300- and SIRT1-related enzyme activity assays, including synthesis and characterization of the conjugated polymer PPE-IDA and design of polylysine peptide substrates.
- Characterized polymer-peptide interactions and optimized probe composition and enzymatic reaction conditions to establish a fluorescence-based workflow for enzyme activity detection.

Undergraduate Researcher, SUSTech – China

2017-2018

- Developed and evaluated asymmetric annulation and catalyst-free cycloaddition reactions. Performed small-molecule synthesis, purification, and characterization using glovebox techniques, chromatography, NMR, HPLC, and X-ray crystallography.

TECHNICAL SKILLS

- **Bioanalytical & Biochemical Assays:** PCR, ELISA, enzyme activity assays, fluorescence-based assays, SDS-PAGE, Western blotting, DNA-PAGE
- **Molecular Biology & Sample Preparation:** Oligonucleotide and primer design, cell homogenization, nucleic acid/protein extraction and quantification, protein expression and purification, dialysis, membrane protein reconstitution
- **Analytical Characterization:** mass spectrometry, HPLC-MS, mass photometry, HS-AFM, TEM, NMR, CD, fluorescence and UV-Vis spectroscopy, DLS
- **Data Analysis & Scientific Software:** Python, MATLAB, ImageJ/Fiji, ChimeraX, Igor Pro, Origin, MNovo, ChemDraw

ADDITIONAL EXPERIENCES

Associate Consultant, Texas Venture Lab – Austin, TX

Jan 2025-May 2025

- Conducted market research, competitive analysis, and financial modeling for a biotechnology startup developing gene-editing approaches for mosquito population control.
- Presented technical and commercial findings to startup stakeholders and translated scientific data into actionable recommendations.

Teaching Assistant, UT Austin – Austin, TX

2022-Present

- Led discussion or laboratory sessions and explained chemistry concepts to undergraduate students.
- Evaluated assignments and provided feedback to students from diverse academic backgrounds.

HS-AFM Instrument Manager, Lin Bio-AFM Lab – Austin, TX

2022-Present

- Managed shared HS-AFM instrumentation, trained new users, troubleshoot instrument performance and experimental workflows, and developed standardized protocols for biological sample preparation and imaging.

SELECTED PUBLICATIONS & PRESENTATIONS

- **Zhang, P.;** Guo, M.; Zhang, Y.; Lambowitz, A.; Lin, Y., Real-Time Visualization of G2L4 Reverse Transcriptase in DNA Repair via Microhomology-Mediated End Joining. Manuscript accepted at *Nat. Comm.*
- **Zhang, P.;** Tan, C., Cross-Reactive Fluorescent Sensor Array for Discrimination of Amyloid Beta Aggregates. *Anal. Chem.* **2022**.
- Qian, C.*; **Zhang, P.*;** Li, W.; Li, P., Phosphine Catalyzed Enantioselective [1 + 4]-Annulation of Morita-Baylis-Hillman Carbonates with α,β -Unsaturated Imines. *Asian J. Org. Chem.* **2019**. (*Co-first author)
- **Zhang, P.;** et al., Real-Time Visualization of Intermolecular Interactions between SARS-CoV-2 Spike Protein and NTD-Binding Antibody Using High-Speed Atomic Force Microscopy. *Poster presentation, Biophysical Society Annual Meeting, 2024*.

SELECTED SCHOLARSHIPS & AWARDS

Chemistry Department Research Fellowship

UT Austin, 2026

Outstanding M.Sc. Thesis Award and Outstanding Graduate

Tsinghua University, 2022

Outstanding B.Sc. Thesis Award and Outstanding Graduate

SUSTech, 2019